

Love the one you're with: genomic evidence of panmixia in the sablefish (*Anoplopoma fimbria*)

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Abstract: Understanding the genetic structure of a fishery may help delineate stocks and is directly applicable to resource management. To date, studies have not found clear population genetic structure across the range of the sablefish (*Anoplopoma fimbria*), yet significant biological differences are recognized. Here we use restriction site-associated DNA sequencing to develop thousands of single nucleotide polymorphisms (SNPs) throughout the sablefish genome and assess population genetic structure and examine the genome for SNPs under natural selection. Our study was unable to target spawning groups, having the potential to bias analyses that require a priori hypotheses of population structure. Low and insignificant levels of differentiation ($F_{ST} = 0.0002$) were observed among survey areas, and analyses of population structure suggested a single population. Only two SNPs were significantly associated with environmental variables. These results are likely due to considerable population mixing and suggest a single panmictic group of sablefish off the west coast of the United States and Alaska that is likely a consequence of a complex juvenile life history and long range movements as adults.

Résumé : La compréhension de la structure génétique d'une pêche pourrait aider à en délimiter les stocks et est directement applicable à la gestion des ressources. À ce jour, les études n'ont pas relevé de structure génétique claire des populations à l'échelle de l'aire de répartition de la morue charbonnière (*Anoplopoma fimbria*), même si des différences biologiques significatives ont été reconnues. Nous utilisons le séquençage d'ADN associé à un site de restriction pour produire des milliers de polymorphismes mononucléotidiques (SNPs) à la grandeur du génome de la morue charbonnière et évaluons la structure génétique des populations et examinons le génome pour déceler des SNP qui reflèteraient une sélection naturelle. L'étude n'a pu cibler des groupes reproducteurs, ce qui pourrait biaiser les analyses qui requièrent des hypothèses a priori concernant la structure des populations. Des degrés de différenciation faibles et non significatifs ($F_{ST} = 0,0002$) ont été observés parmi les régions évaluées, et les analyses de la structure des populations indiqueraient une population unique. Seuls deux SNP sont associés significativement à des variables environnementales. Ces résultats découlent vraisemblablement d'un important mélange des populations et indiqueraient qu'il n'y a qu'un groupe panmictique de morues charbonnières au large de la côte Ouest des États-Unis et de l'Alaska, la conséquence probable du cycle biologique complexe des juvéniles et des déplacements sur de longues distances des adultes. [Traduit par la Rédaction]

Introduction

The concept of a stock is central to the conservation and management of marine fisheries (Begg and Waldman 1999). Under the stock concept, managers define semi-discrete boundaries to fish populations to make decisions regarding those management units (Begg et al. 1999). Knowledge of genetic structure or an understanding of localized adaptation may help delineate fish stocks and can be extremely relevant in the management of a fishery (Hauser and Carvalho 2008). Many commercially important marine species fall into the category of "classical marine species", characterized by large geographic distributions and population sizes, high fecundity, and pelagic larvae (Nielsen and Kenchington 2001). Many of these species typically display relatively low levels of genetic differentiation among population groupings; however, significant genetic differentiation has been observed in a number of these species, even at relatively small spatial scales (Nielsen and Kenchington 2001; Nielsen et al. 2009).

The sablefish (*Anoplopoma fimbria*) could be considered a "classical marine species" (see Nielsen and Kenchington 2001). It is a long-lived, demersal species found in the northern Pacific Ocean, where it inhabits deep water along the continental slope (Sasaki 1985; Beamish and McFarlane 1988; Beamish et al. 2006). It is found along the west coast of North America from Baja California to the Bering Sea, throughout the Aleutian Islands into waters off the Kamchatka Peninsula, Russia, and northern Japan (Sasaki 1985). It is generally accepted that two sablefish stocks exist in the North-east Pacific Ocean (Hanselman et al. 2012). A northern stock inhabits waters off of the Alaskan Coast and northern British Columbia, while a southern stock is found off southern British Columbia and the west coast of the contiguous United States (Hanselman et al. 2012). The division of these stocks is primarily based on tagging studies and differences in growth and maturity between fish in these areas (Kimura et al. 1998; Hanselman et al. 2012; Head et al. 2014). Sablefish in the northern stock generally grow to and ma-

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ture at larger sizes than the fish in the southern stock (Sasaki 1985; Kimura et al. 1993; Head et al. 2014).

The sablefish is a highly mobile species, and there is little known about the timing or location of spawning. It is thought that spawning occurs at depth along the continental slope (Mason et al. 1983), though no specific spawning sites have been documented in the literature. The eggs are pelagic and believed to incubate at depths of about 400 m and descend to depths of 1000 m or more after hatching (Alderdice et al. 1988; McFarlane and Beamish 1992). During yolk sac absorption, the larvae ascend to the surface where they are very abundant in the neuston (Doyle 1992). The development of eggs to 40 mm larvae is estimated to take about 10 weeks in the wild (Boehlert and Yoklavich 1985; Kendall and Matarese 1987). Juvenile sablefish utilize nearshore areas (1–2 years) before they begin to gradually move back out to the slope as they reach maturity (Maloney and Sigler 2008).

While the current paradigm suggests that there are two biological stocks of sablefish in the Northeast Pacific Ocean, studies have found evidence of biological differences on a finer geographic scale (Hanselman et al. 2012). For example, stable isotope analysis of otoliths from juvenile sablefish suggests that three spawning stocks may exist from Oregon to Vancouver Island, British Columbia (Gao et al. 2004). Differences in maturity and growth parameters observed between groups of sablefish collected off the west coast of the United States might suggest population differences occur within the southern stock (Head et al. 2014). It has also been suggested that genetic differences may be responsible for the so-called “soft flesh problem” observed in sablefish, where fish caught in deeper water generally have softer flesh, less protein content, increased lipid levels, and have a less desirable flesh quality when compared with fish harvested from shallower water (Norris et al. 1987; Karinen et al. 2010).

Previous genetic studies have attempted to characterize the population structure of this species (Tsuyuki and Roberts 1969; Gharrett et al. 1982; Tripp-Valdez et al. 2012). These studies suggest that there is little or no genetic structure across large geographic scales; however, they have only examined variation at genetic markers that are presumed to be neutral to the effects of natural selection, such as microsatellites. While useful in examining demographic processes, care should be taken when interpreting low or insignificant values of genetic differentiation derived from neutral markers in marine populations, as these estimates may not accurately represent contemporary patterns of stock structure in such large populations (Allendorf and Phelps 1981; Conover et al. 2006). Large population sizes typical of many marine species make them less susceptible to the effects of genetic drift; therefore, apparent genetic homogeneity may persist at neutral genes even at low levels of migration (Allendorf and Phelps 1981; Nielsen et al. 2009). Given that large populations are less susceptible to drift, beneficial alleles are less likely to be lost through drift and may respond to relatively weak selection pressure in marine fish species (Nielsen et al. 2009). Also, many more migrants are needed to effectively counteract the effects of natural selection than are required to maintain genetic homogeneity in the face of genetic drift (Conover et al. 2006). So, while neutral markers may fail to show any patterns of significant differentiation, significant adaptive variation may persist (Conover et al. 2006; Larsen et al. 2007; Hauser and Carvalho 2008). In species such as the sablefish that have large geographic ranges and large population sizes, apparent genetic homogeneity may be observed as a result of low levels of genetic drift even when migration rates are low. In this scenario, genes subject to natural selection may offer more power to distinguish stocks over loci that are presumed to be selectively neutral, and by incorporating non-neutral genes it may be possible to more accurately assess the genetic structure of sablefish stocks.

The emerging field of landscape genomics provides a framework to examine the association between environmental variables and genomic variation in wild populations. Schwartz et al.

(2010) refers to landscape genomics as “...the simultaneous study of tens-to-hundreds of markers, ideally including markers in candidate adaptive genes (genes under selection), with georeferenced samples collected across a landscape”. While approaches like this are not particularly abundant in the literature on classical marine fishes, they have been successfully applied to analyses of local adaptation and population structure in other fishes such as salmon (Zueva et al. 2014) and other organisms such as trees (Bashalkhanov et al. 2013; Frichot et al. 2013; Sork et al. 2013), plants (Bothwell et al. 2013; Jones et al. 2013), goats (Pariset et al. 2009), and humans (Frichot et al. 2013).

Here, restriction site-associated DNA sequencing (RAD-seq) was used to obtain single nucleotide polymorphism (SNP) markers that were used to (i) test the null hypothesis of genetic homogeneity of sablefish collections from the Bering Sea, Gulf of Alaska, and the west coast of the US, using traditional population genetic analyses, (ii) investigate the possibility of cryptic population structure using individual-based genetic methods, and (iii) test for signatures of natural selection in the sablefish genome by associating SNP genotypes with environmental conditions. Lack of information on the location and timing of sablefish spawning meant that groups of spawning fish were not specifically targeted during sampling, posing limitations on a priori hypotheses regarding population boundaries. Therefore, this should be considered when interpreting the results.

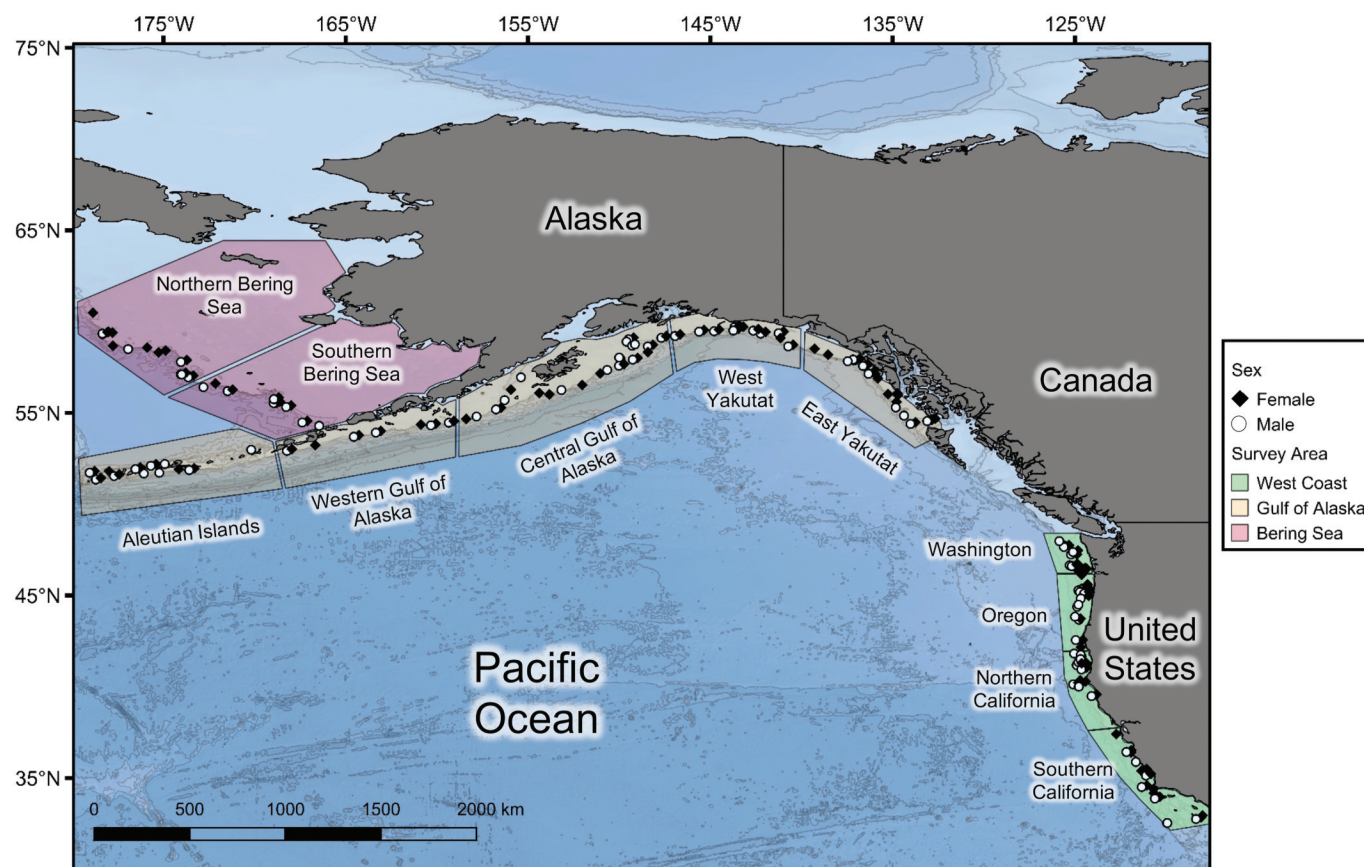
Materials and methods

Sample collection

Fin clips were collected and stored in 95% ethanol as a part of the National Oceanic and Atmospheric Administration's National Marine Fisheries Service (NOAA, NMFS) surveys in the summer of 2012. The Fishery Resource Analysis and Monitoring (FRAM) Division at NOAA's Northwest Fisheries Science Center collected fin clips ($n = 600$) during their annual US West Coast Groundfish Bottom Trawl Survey. As part of their annual longline survey, the Marine Ecology and Stock Assessment (MESA) Division of NOAA's Alaska Fisheries Science Center (AFSC) collected fin clips ($n = 1929$) from the Gulf of Alaska. The Resource Assessment and Conservation Engineering Division (RACE) at AFSC collected fin clips ($n = 140$) from sablefish opportunistically encountered as part of their trawl survey of the Bering Sea Slope. In addition to fin clips, biological data were collected for each sample, including length, mass, sex, and maturity (FRAM and MESA). The three major regions (West Coast, Gulf of Alaska, Bering Sea) covered by the surveys were further divided into 11 subregions to provide a means to choose samples evenly across the study area (Fig. 1).

To restrict the genetic analysis to fish at a similar life stage, we used maturity data to select individuals that were likely to be adults. The data from the macroscopic assessment of maturity status at the time of sampling could not be used because maturity data were not collected on fish caught in the Bering Sea. While a number of studies have been conducted on size at maturity in sablefish, the results are variable, and nearly all of the data are over 25 years old and focus on localized groups of fish off the west coast of the US (Parks and Shaw 1987; Fujiwara and Hankin 1988; Hunter et al. 1989) and British Columbia (Mason et al. 1983; McFarlane and Beamish 1990). While more current estimates of size at maturity are available for females off the west coast of the US, no estimates are available for males (Head et al. 2014). Therefore, to generate a criterion by which mature fish could be subsampled with reasonable confidence over this study area, data on maturity status for the fish collected in the Gulf of Alaska and West Coast were pooled, and logistic regressions were fit separately for males and females to determine the length at 50% probability of

Fig. 1. Map locations of sablefish samples used for RAD sequencing in this study. Survey areas are shaded (red = Bering Sea: northern Bering Sea, southern Bering Sea; yellow = Gulf of Alaska: Aleutian Islands, western Gulf of Alaska, central Gulf of Alaska, West Yakutat, East Yakutat; green = west coast of the US: Washington, Oregon, northern California, southern California). Black diamonds represent female fish, and white circles are males. Image reproduced from the GEBCO world map 2008, www.gebco.net. [Colour online.]



maturity (L_{50} ; refer to online supplementary material¹, Fig. S1). Only fish above the size thresholds set by L_{50} were considered for genetic analysis. A logistic regression model (GLM, R version 3.1.2) was fit to the data to estimate the parameters of the equation where the probability of being mature (P) was calculated by

$$(1) \quad P = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$

where X is length and β_0 and β_1 are regression coefficients (Quinn and Deriso 1999). The inflection point of the L_{50} was calculated by

$$(2) \quad L_{50} = \frac{-\beta_0}{\beta_1}$$

Males and females above the size thresholds (females: 614.8 mm, males: 573.7 mm) were then randomly selected from each subarea to help ensure an even sample distribution across the study area. In areas where an adequate number of individuals above those size thresholds were not available, the next largest individuals were selected. Out of the total number of fin clips ($n = 2669$), 441 (West Coast: 144, Gulf of Alaska: 224, Bering Sea: 73) were selected for RAD-seq.

Genomic library construction

Genomic DNA was extracted using Qiagen's DNeasy Blood and Tissue Kit (Qiagen, Valencia, California). DNA was quantified using the Quant-iT PicoGreen dsDNA Assay Kit (Invitrogen, Carlsbad, California), and concentrations were adjusted to $12.5 \text{ ng} \cdot \mu\text{L}^{-1}$. Libraries were prepared as described in Miller et al. (2007) with modifications described below. DNA from each sample (500 ng) was digested with either *Sbf*I or *Pst*I restriction endonucleases (New England Biolabs, Ipswich, Massachusetts). Since the *Pst*I recognition site is part of the *Sbf*I recognition site, samples digested with *Pst*I contained all of the *Sbf*I loci, making them suitable for inclusion in the analysis. Barcoded P1 adapters unique to each individual and complementary to the restriction enzyme cut site were ligated onto the digested DNA of each individual. For single end sequencing runs, barcoded, digested DNA was then pooled into libraries containing DNA from 10 (*Pst*I digestions) or 12 (*Sbf*I digestions) individuals for single end sequencing, and a library of four individuals was prepared for paired end sequencing. DNA was then randomly sheared with a Q-sonica Q800R DNA Sonicator (Q-sonica, LLC, Newtown, Connecticut). Gel electrophoresis was used to identify and cut out DNA fragments that were 300–500 bp in length for single end libraries and 200–400, 400–600, 600–900 bp for paired end libraries. Illumina P2 adapters were then blunt-end ligated to the other end of the DNA fragments. Libraries were quantified via qPCR using Kapa Biosystems, Kapa Illumina

¹Supplementary data are available with the article through the journal Web site at <http://nrcresearchpress.com/doi/suppl/10.1139/cjfas-2016-0012>.

library quantification kit (Kapa Biosystems Inc., Woburn, Massachusetts, USA) on an ABI 7900 real time PCR system (Life Technologies, Carlsbad, California). Four libraries were pooled in equimolar concentrations prior to sequencing.

Single end sequencing libraries consisting of samples digested with the restriction endonuclease *Sbf*I were sequenced on an Illumina HiSeq 2500 (Illumina, San Diego, California) for read lengths of 100 bp. Individuals were pooled into a single lane of 48 samples for the single end runs. Paired end sequencing was done on a library of four individuals digested with *Pst*I on the Illumina HiSeq 2500 for read lengths of 100 bp in a single lane and a single library of 10 individuals digested with *Pst*I on an Illumina MiSeq for read lengths of 250 bp.

Processing sequence reads

FastQC (Andrews 2012) was used to assess the quality of each Illumina sequencing run. The raw sequence reads were de-multiplexed, quality-filtered, and trimmed using process_radtags in Stacks (version 1.21) (Catchen et al. 2011, 2013). The sequence reads were trimmed to 75 bp ($-t$ 75) during de-multiplexing to remove poor-quality base calls at the end of each sequence read. Reads with low quality scores ($-q$) and any uncalled bases were removed ($-c$). The default sliding window parameter ($-w$) of 0.15 was used in process_radtags when evaluating read quality scores. Bowtie (version 1.0.0) (Langmead et al. 2009) was used to align sequence reads from each individual to the sablefish draft genome (GenBank: AWGY00000000.1) and mitogenome (NCBI RefSeq: NC_018119.1) (Rondeau et al. 2013). Up to three mismatches were allowed when aligning reads to the genome, and reads with more than one valid alignment were discarded. The Stacks program pstacks was used to form loci from stacks of sequence reads within individuals. A minimum of five reads was required to form a stack at a locus. The bounded SNP calling model implemented in pstacks was used (upper bound: 0.1, lower bound: 0.001, α : 0.05). Eleven individuals were then selected at random from each of the 11 geographic subregions (Fig. 1) to create a catalog of reference loci in the Stacks program cstacks. The Stacks program sstacks was then used to match loci in each individual back to the catalog of reference loci. The program populations was used to export a variant call format (vcf) file (Danecek et al. 2011) containing the genotype information for all individuals. The program populations was used to process and export RAD tags within individuals that have minimum stack depth of eight reads for further analyses.

SNP quality filtering

First, to restrict the analyses to nuclear SNPs, all RAD loci in the mitochondrial genome were removed from further analysis. Next, RAD tags were culled from further filtering if they contained more than three SNPs. The program VCFtools (version 0.1.12) (Danecek et al. 2011) was then used to filter out loci with a global minor allele frequency less than 0.05 and with more than 20% missing data. Individuals were then removed from further analyses if they were missing genotypes at more than 40% of the filtered loci. For tags that contained more than one SNP, only the SNP with the highest minor allele frequency was retained. Linkage disequilibrium was assessed by calculating the r^2 statistic between each pair of SNPs using the program Plink (version 1.07; Purcell et al. 2007). In cases where $r^2 > 0.8$, the SNP with more missing data was culled from the data set. Exact tests for Hardy-Weinberg equilibrium (HWE) were conducted in VCFtools, and SNPs that were out of HWE in any one population (West Coast, Bering Sea, Gulf of Alaska) were removed in all populations prior to conducting analyses that required population groups to be defined a priori.

Population structure

To assess the genetic structure of sablefish across their sampled range, we combined the use of traditional population genetic techniques with individual-based analyses that may be better suited

when individuals are sampled across a landscape (Jones et al. 2013). We pooled individuals into population groups defined by the survey on which they were collected (West Coast: FRAM; Gulf of Alaska: MESA; Bering Sea: RACE). Levels of expected heterozygosity were calculated in VCFtools for each population, and F_{IS} was calculated for each SNP both globally and in each survey area. Fisher's exact test as implemented in Genepop version 4.2.2 (Rousset 2008) was used to test for differences in allele frequency between survey areas. Markov chain parameters of 10 000 dememorization steps, 100 batches, and 5000 iterations per batch were used for the exact test in Genepop. POWSIM (Ryman and Palm 2006) was used to assess the statistical power of this data set to detect differences in allele frequency between survey areas at known levels of genetic divergence or F_{ST} for both the global and pairwise tests. Twenty-five replicate runs were used with the same Markov chain parameters as Genepop. The number of samples selected from each population at the end of the drift simulations was defined by the sample sizes of each area in the filtered SNP data set used for this study. A version of POWSIM was specifically compiled to accommodate the size of this data set (Ryman and Palm 2006). Genepop was also used to test for differences in genotype frequency between the survey areas using the same Markov chain parameters used to test for differences in allele frequency. F_{ST} (Weir and Cockerham 1984) was estimated globally and between all group pairs using Genepop. The R package StAMPP (Pembleton et al. 2013) was used to test for significance of the pairwise F_{ST} values obtained in Genepop. StAMPP was chosen because it also estimates F_{ST} using the method of Weir and Cockerham (1984) and is comparable to Genepop. Individuals were randomized among groups for 1000 permutations, and a p value was obtained by converting the simulated values to an object of the class randtest in the R package ade4 (Chessel et al. 2004).

Isolation by distance (IBD) was evaluated by calculating Rousset's $\hat{a}$ (Rousset 2000) between individuals in SPAGeDi version 1.4 (Hardy and Vekemans 2002). Geographic distance between individuals was calculated as great-circle distance in the fields package in R (Nychka et al. 2014). Distance between individuals along the 500 m isobath was also estimated using the R package gdistance to route individuals between sample locations along the 500 m isobath (van Etten 2014). Significance of IBD relationships were assessed by Mantel's test with 1000 permutations implemented in R using the vegan package (Oksanen et al. 2013). Values of great-circle distance were log-transformed prior to analysis to account for the possibility of dispersal in two dimensions, while distances along the 500 m isobath were not because under that dispersal scenario, it is assumed to take place in one dimension (Rousset 1997).

To investigate the possibility of cryptic population genetic structure, we used a combination of methods that do not rely on predefined population groupings. First, the program SmartPCA in the software package Eigensoft version 5.0.2 was used to perform principal components analysis (PCA) of the genotype data (Patterson et al. 2006). Default outlier removal parameters were used in SmartPCA ($> |6\sigma|$ on one of the top 10 PCs, 10 iterations). PCA was used because it does not rely on any underlying population genetic models and does not attempt to group individuals into discrete populations (Patterson et al. 2006). Individuals are placed along axes of variation (PCs), and once plotted, a decomposition of the genetic structure from the entire set of SNPs can be visualized (Patterson et al. 2006). The spatial clustering algorithm implemented in BAPS version 6 was also used to examine population genetic structure (Corander et al. 2008). Unlike PCA, BAPS uses a model that attempts to minimize linkage and Hardy-Weinberg disequilibrium within groups of samples while also incorporating spatial information to improve the partitioning of individuals into biologically meaningful groups (Latch et al. 2006; Corander et al. 2008). Partitions of one to 30 groups were tested in BAPS.

Table 1. Table of F_{ST} estimates for each pair of survey groups.

	Bering Sea	Gulf of Alaska	West Coast
Bering Sea	—	0.1199	0.0889
Gulf of Alaska	0.0002	—	0.0589
West Coast	0.0002	0.0002	—

Note: F_{ST} is shown below diagonal, and p values are shown above.

Landscape genomics

Environmental variables

The samples obtained for this study were collected using bottom contact fishing methods; thus, depth of capture for each individual was extracted from the gebco_08 30 arc second grid bathymetry data (IOC, IHO, and BDOC 2003) using the GPS location of each individual in QGIS (QGIS Development Team 2009). Environmental conditions at the sampling location of each individual were estimated using salinity and temperature profiles collected by the Argo profiling float network. These data were collected and are made freely available by the International Argo Program and the national programs that contribute to it (<http://www.argo.ucsd.edu>; <http://argo.jcommops.org>). The Argo Program is part of the Global Ocean Observing System. Argo data from the time period that the samples were collected (1 March 2012 – 31 October 2012) were obtained from the Coriolis project's web-server (<http://www.coriolis.eu.org>). Temperature and salinity were estimated using the sampling location and bottom depth for each individual using the computer program Ocean Data View version 4.6.2 (Schlitzer 2014). To reduce collinearity among the environmental variables, Pearson's correlation coefficient (r) was calculated between each pair of variables in R (R Core Team 2013), and if a pair of variables were correlated with an $|r| > 0.8$, one variable was removed from the association analyses.

Association analyses

Latent factor mixed models implemented in LFMM (Frichot et al. 2013) were used to test for genotype–environment associations. LFMM has been shown to reduce the number of false positives in association studies by accounting for population history, including patterns of IBD (Jones et al. 2013). Population structure is modeled using unobserved variables or “latent factors” (Frichot et al. 2013). We chose to use a single latent factor (k) because our analyses suggested the existence of a single cluster. LFMM's Gibbs sampler was set to estimate model parameters using 10 000 sampling iterations with 1000 iterations discarded as burn-in. The LFMM results did not change over replicate runs for the same variable, so we determined that the Gibbs sampler was run long enough. Significance (p values) in LFMM was adjusted for a false discovery rate of 0.05 by calculating q values (Storey and Tibshirani 2003) in the R package q -value (Dabney and Storey 2014).

Gene annotation

SNPs with a significant association to an environmental variable were annotated by using Ensembl's BLASTN search tool (http://www.ensembl.org/Gasterosteus_aculeatus/Tools/Blast?db=core) to align the full contig sequence from the sablefish genome assembly (Rondeau et al. 2013) containing the SNP tag of interest to release 78 of the stickleback (*Gasterosteus aculeatus*) BROAD S1 genome (Kersey et al. 2014). An E value of $<1e^{-03}$ was used as a cutoff to identify valid alignments. The alignments were then viewed using the BROAD S1 genome browser in Ensembl (Kersey et al. 2014). While the raw sequence reads were aligned to contigs from the sablefish draft genome, those contigs are not annotated. The stickleback genome was chosen because it is fully annotated, and stickleback are the most closely related species to the sablefish for which a complete genome is available (Rondeau et al. 2013). It is thought that the divergence of sablefish and stickleback lineages occurred less than 150 million years ago and possibly as recent as 50 million years ago (Rondeau et al. 2013). A high degree of syn-

Table 2. Significance (p values; below diagonal) for tests of allelic differentiation between each population pair and statistical power (above diagonal) of each comparison calculated in POWSIM for the level of F_{ST} observed between each population pair (Table 1).

	Bering Sea	Gulf of Alaska	West Coast
Bering Sea	—	0.0000	0.0000
Gulf of Alaska	0.9997	—	0.0000
West Coast	0.9970	0.8230	—

teny between stickleback and sablefish chromosomes is also observed (Rondeau et al. 2013).

Results

Sequencing

There were on average 145 960 310 reads per lane for the Illumina HiSeq runs. On average, we were able to recover 80% of the barcoded reads per lane with 3 274 890 reads per individual. On average, 65% of the reads for each individual aligned to the draft genome unambiguously, 16% of the reads had more than one valid alignment and were discarded, and 19% failed to align altogether. The Stacks pipeline produced 100 317 SNPs in 29 242 tags, and after SNP quality filtering (see Materials and methods) in VCFtools and Plink, the final data set consisted of 2661 SNPs in 404 individuals (West Coast: 117; Gulf of Alaska: 217; Bering Sea: 70). While 441 individuals were submitted for sequencing, 37 were removed because of missing data, leaving 404 individuals for subsequent analyses.

Population structure

For analyses that required a priori population groupings (West Coast, Gulf of Alaska, Bering Sea), a number of SNP loci (624) were found to be out of HWE and were filtered prior to any population genetic analyses, for a total of 2037 SNPs. The removal of SNPs from the data set that were out of HWE brought the mean F_{IS} across loci closer to 0 both globally and for all population groups (Table S1¹; Fig. S2¹). Differentiation among groups was very low (global $F_{ST} = 0.0002$). Pairwise estimates of F_{ST} between the three survey areas were all 0.0002 (Table 1) and nonsignificant. No significant differences in genotype frequencies were detected between any population pairs (BS–GOA: $p = 1.0$; GOA–WC: $p = 0.999$; BS–WC: $p = 0.889$). The simulations conducted in POWSIM revealed that statistical power was not sufficient at the levels of differentiation (F_{ST}) observed to identify differences in allele frequency between population pairs using Fisher's exact test (Table 2).

A small number of individuals had very low estimates of genetic distance ($\hat{a} < -0.4$) between them (Figs. 2a and 2b). These were considered to be outliers, and all comparisons with those samples were removed from the analysis. A number of samples were collected within 1 km from another sample, and multiple samples were collected at many of the sampling stations yielding a distance of 0 km for those comparisons. To successfully log-transform distance measures of 0 km and produce positive values for comparisons of samples collected within 1 km of each other, a constant of 1 km was added to each measurement of great-circle distance. The relationship between genetic ($\hat{a}$) and geographic distance between individuals was nonsignificant for all tests prior to outlier removal (log(great-circle + 1 km); Mantel's $R = 0.007$, $p = 0.181$, distance along 500 m isobath; Mantel's $R = -0.016$, $p = 0.846$) and after (log(great-circle + 1 km); Mantel's $R = 0.005$, $p = 0.293$, distance along 500 m isobath; Mantel's $R = -0.016$, $p = 0.847$), suggesting that there is no pattern of IBD in sablefish (Fig. 2).

Results from the Bayesian clustering analysis and PCA did not identify genetic groups defined by geographic region or reveal any cryptic patterns of population genetic structure. Seven samples were more than six standard deviations along one of the first 10 PCs and flagged as outliers by SmartPCA. These samples were excluded

Fig. 2. Patterns of isolation by distance for sablefish in the Northeast Pacific Ocean. Pairwise genetic distance between individuals ($\hat{a}$) is plotted against geographic distance calculated as log of great-circle distance + 1 km (a) and distance along the 500 m isobath (b) before removing individuals identified as outliers. Relationship between log of great-circle distance + 1 km (c) and distance along the 500 m isobath (d) after outliers were removed.

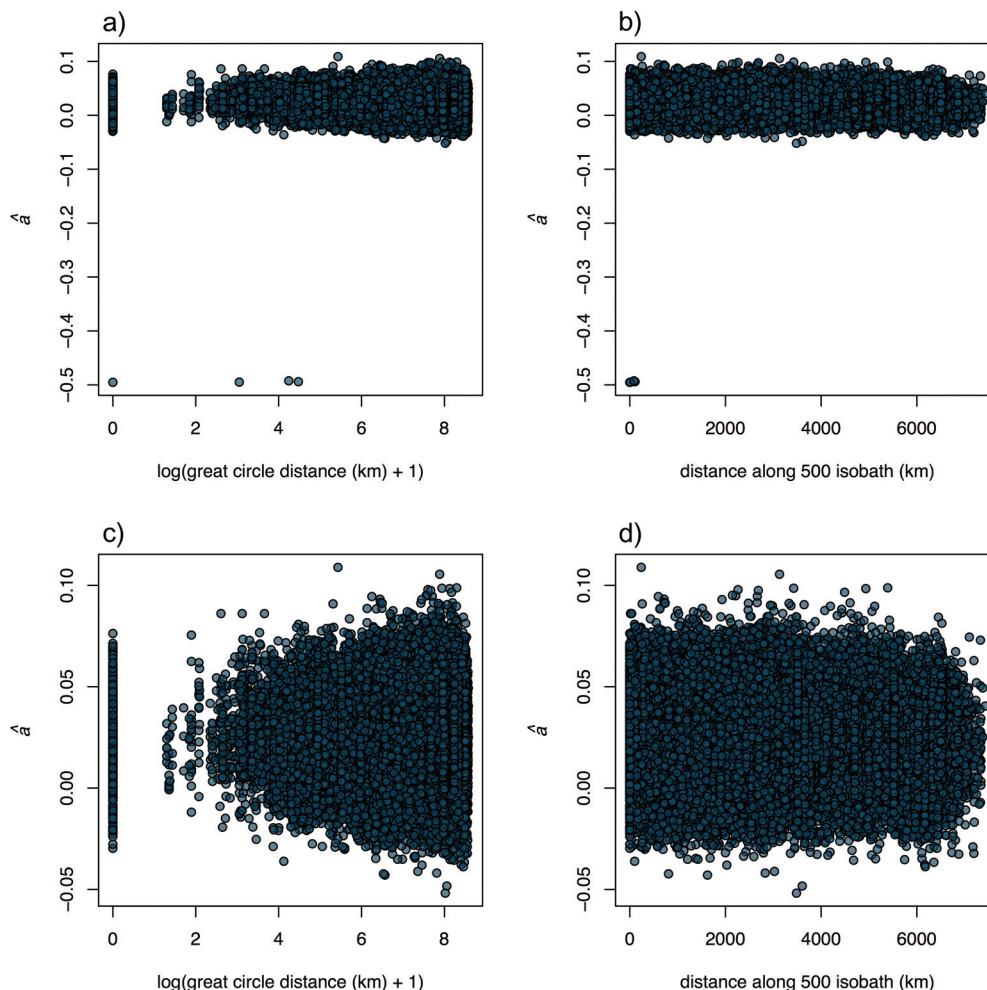


Fig. 3. Plot of the first two principal component axes in a principal component analysis of 2661 RAD-seq derived SNPs. Points are colored by the area (yellow square = Bering Sea, red triangle = Gulf of Alaska, blue square = West Coast) that each individual was collected. Lines represent 95% confidence ellipses calculated using the R package *vegan* (Oksanen et al. 2013) for each group. [Colour online.]

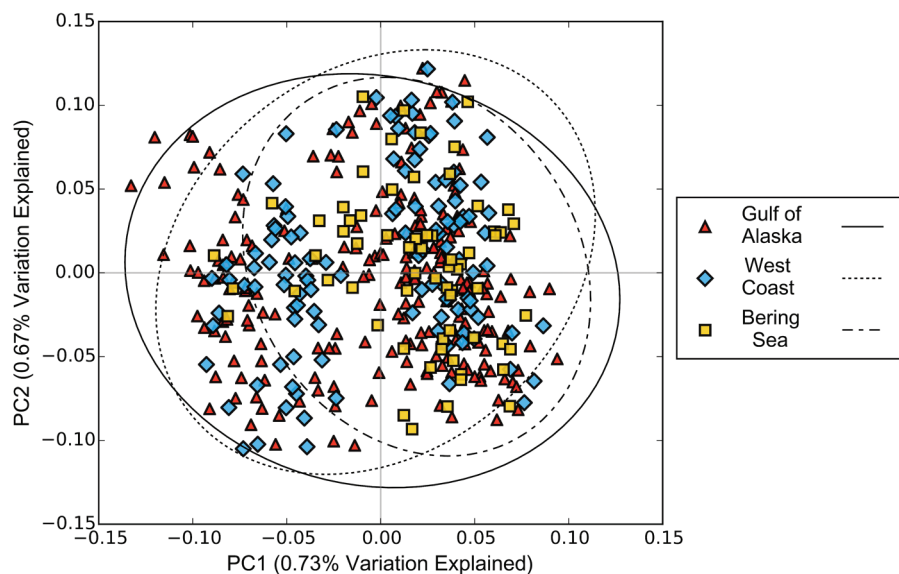


Table 3. Significant SNP–environment associations found using LFMM.

RAD tag	Contig	Base pair position	Variable	<i>p</i> value	<i>q</i> value
9124	gi 559591484 gb AWGY01151885.1	11333	Depth	6.35E-06	0.0338
12048	gi 559609892 gb AWGY01133477.1	4995	Depth	2.81E-08	0.0003

Note: *q* values were calculated from *p* values using a false discovery rate threshold of 0.05.

from the PCA as a result. The first PC explains a very small proportion of the total variation (0.73%). When individuals were plotted on the first two PC axes calculated from the SNP data set in SmartPCA, there was considerable overlap among the sampling regions, suggestive of little to no differentiation between the survey areas (Fig. 3). There was no clear geographic pattern to the placement of individuals along the PCA axes. Similarly, a single group was also identified by BAPS.

Environmental associations

Of the five environmental variables selected for this analysis, salinity and depth were correlated with one another (Pearson's $r > |0.8|$). Salinity was subsequently excluded from further analyses because it was estimated from the Argo float profile data and not directly measured. After *p* values were adjusted for multiple testing, the latent factor mixed models implemented in LFMM detected two significant SNP associations with depth (Table 3). These SNPs were contained in RAD tags 9124 and 12048 (Table 3).

Gene annotation

Each of the sablefish genome contigs containing the SNPs associated with depth in this study were successfully aligned to the stickleback genome. The sablefish contig 152341 (GenBank: AWGY01151885.1) containing the SNP in RAD tag 9124, aligned to the stickleback linkage groupXX (Table S2¹). The contig did not align continuously to this region, but many fragments aligned in order with gaps between them. While the portion of the contig containing the SNP (bp 11 333) did not align directly to the stickleback genome, based on the order of the contig segments that did align, the SNP is presumably part of an intron of the ETS translocation variant 1 (*etv1*) gene (Ensembl ID: ENSGACG00000012802.1). The alignment was similar for sablefish contig 133848 (GenBank: AWGY01133477.1). A number of fragments aligned to stickleback linkage groupVI (Table S3¹). Based on the alignment order of the fragments, this SNP (bp 4995) occurs in an intron of the gene that codes for the FAM196ab protein (*fam196ab-201*) (Ensembl ID: ENSGACG00000009611.1).

Discussion

Using thousands of SNP markers throughout the sablefish genome, we were unable to find evidence that would suggest there are genetically distinct populations of sablefish within their North American range. While the current paradigm for sablefish stock structure suggests the existence of northern and southern biological stocks, results from population genetic and individual-based genetic analyses in this study are suggestive of a single genetic group across the North American range of the sablefish. This suggests random mating of individuals across the sablefish range, and this pattern of population genetic structure is often referred to as panmixia. The estimates of F_{ST} observed in this study are well below the level required to maintain similar allele frequencies between populations ($F_{ST} < 0.02$; Lowe and Allendorf 2010). Further results from our analysis of IBD and individual-based genetic analyses were also suggestive of panmixia. A nonsignificant relationship was observed between genetic and geographic distance, and genetic groupings were not identified by the PCA or BAPS, suggesting that dispersal may occur over large spatial scales or gene flow occurs between distant areas. Also, very few SNPs were associated with the environmental variables examined in this study. While this suggests that there is essentially no selective

pressure associated with geography or the environmental conditions used in this study, it could also be an artifact of high levels of gene flow throughout the Northeast Pacific that homogenizes adaptive variation brought about by selection at earlier life stages.

The extensive movements that characterize sablefish life history offer one explanation for the apparent genetic homogeneity and lack of adaptive variation observed in this study. Sablefish undergo a pelagic larval period and utilize nearshore areas as juveniles for a number of years before they move back to deep water habitat on the continental slope (Sasaki 1985; Maloney and Sigler 2008). To our knowledge there are no studies that have examined the capacity for natal homing in this species, so the location where an individual migrates back to the slope could be very far from where it was conceived. Tagging data also demonstrates that both juvenile and adult sablefish (McFarlane and Saunders 1993; Maloney and Sigler 2008) have the capacity for frequent long-distance movements, often moving over 200 km in a single year (Kimura et al. 1998; Morita et al. 2012; Hanselman et al. 2015). This, coupled with relatively long lifespans where individuals can exceed 40 years old (Kimura et al. 1993; Maloney 2004; Head et al. 2014), provides ample opportunities for gene flow to occur.

While the lack of significant SNP–environment associations could indicate that the environmental conditions examined in this study have little selective influence on the genome in adult sablefish, it may also provide further support for the idea that extensive gene flow is occurring. As migration and subsequent gene flow occur, potentially adaptive alleles are transferred among populations or geographically distant areas. In cases where migration rate is large relative to the strength of selection and the strength of selection for alternate alleles varies spatially, locally adaptive alleles are easily lost. Often, alleles with the best average reproductive success become fixed in the metapopulation, preventing local adaptation (Lenormand 2002). This phenomenon is termed gene-swamping (Lenormand 2002) or patch-swamping (Felsenstein 2015). Movement throughout the entire life cycle of the sablefish must be considered when interpreting these results. For example, if the environment imposes natural selection at earlier life stages (e.g., during larval development, recruitment, or when juvenile sablefish are in nearshore areas), environmental conditions associated with the sampling location of adult fish may not capture this selective signal, especially if migration and dispersal distances are long.

In the case of the sablefish, regional differences in life history (growth and maturity) have been clearly demonstrated by a number of studies (Kimura et al. 1993, 1998; Echave et al. 2012; Head et al. 2014). While previous genetics studies, including the data presented here, suggest a lack of genetic structure (Gharrett et al. 1982; Tripp-Valdez et al. 2012; Orlova et al. 2014), an interesting question that arises is how can regional biological differences be observed when migration rates are high enough that genetic variation is effectively homogenized? Phenotypic plasticity offers one explanation for this phenomenon. Phenotypic plasticity is the ability for similar genotypes to display different phenotypes in response to variable environmental conditions. Latitudinal variation in growth and size at first sexual maturity has been documented in other marine fish species in the Northeast Pacific Ocean as well. Pacific cod (*Gadus macrocephalus*) (Ormseth and Norcross 2009), dover sole (*Microstomus pacificus*) (Abookire and

Macewicz 2003), and a number of rockfish species (*Sebastes* spp.) (Haldorson and Love 1991; Gertseva et al. 2010) all exhibit some degree of latitudinal variation in growth and size at maturity. While some of these species are not necessarily considered panmictic (Stepien 1999; Cunningham et al. 2009; Spies 2012), the observation of geographic variation in life history parameters when genetic evidence supports panmixia is not unique to sablefish. Genetic evidence suggests a single coastal population of yelloweye rockfish (*Sebastes ruberrimus*) from Alaska to Oregon (Siegle et al. 2013), yet growth and maturity parameters vary within this area (Wallace 2001; O'Connell et al. 2003). This pattern has been observed in commercially important marine fishes in other oceans as well, including Greenland halibut (*Reinhardtius hippoglossoides*) in the Northwest Atlantic (Bowering 1983; Morgan and Bowering 1997; Roy et al. 2014), orange roughy (*Hoplostethus atlanticus*) within different ocean basins though out their range (Branch 2001; White et al. 2009; Varela et al. 2013), and red snapper (*Lutjanus campechanus*) in the Gulf of Mexico (Gold et al. 2001; Fischer et al. 2004; Saari et al. 2014).

Phenotypic plasticity may help explain the differences in growth and maturity observed in sablefish stocks across their range. In the absence of data that would suggest there is a genetic component to the regional differences in life history parameters observed in sablefish, it is certainly possible that environmental conditions cause these differences. Laboratory studies suggest that sablefish growth is tightly coupled to environmental conditions where physiological limits to maximum growth are often imposed by temperature or available food sources (Sogard and Olla 2001, 2002; Sogard and Spencer 2004). The differences in oceanographic conditions between the west coast of the US and Alaska (Huyer 1983; Ware and Thomson 2005) may influence ecosystem productivity and the quantity and quality of the food that is available to young sablefish in different regions (Hooff and Peterson 2006; Kudela et al. 2008). In the wild, Sogard (2011) found that growth is negatively correlated with the Pacific Decadal Oscillation index, suggesting that cooler conditions actually improve growth. After adjusting growth rate for temperature and size, juvenile sablefish actually tended to grow better during cooler years (Sogard 2011). This is likely a function of better quality food sources and improved foraging opportunities provided by cooler conditions during negative Pacific Decadal Oscillation regimes (Hooff and Peterson 2006; Sogard 2011). While these studies focus on the southern sablefish stock, Alaskan waters (Stabeno et al. 2004; Ware and Thomson 2005) and those off Washington and Oregon (Hooff and Peterson 2006; Kudela et al. 2008) may be more productive in general and support the high levels of consumption required for the fast growth of juvenile sablefish. Juvenile sablefish in St. John Baptist Bay, Alaska, have been observed foraging on salmon carcasses, and this seasonally abundant and high-energy food source may improve sablefish growth in these areas (Coutré et al. 2015). The regional differences in environmental conditions experienced by sablefish especially at early life stages may alter the rate of adult growth and timing of first maturation (Berrigan and Charnov 1994). More surplus energy may be available for growth at higher latitudes, and this could be why the regional differences in sablefish growth and maturity exist in the absence of genetic differentiation.

In summary, the lack of genetic structure observed in this study supports past genetic work and is not surprising when viewed in light of tagging data that suggests that these fish move frequently and over long distances (Kimura et al. 1998; Maloney and Sigler 2008; Morita et al. 2012; Hanselman et al. 2015). High annual movement probabilities (Kimura et al. 1998; Morita et al. 2012; Hanselman et al. 2015) combined with a relatively long lifespan (Kimura et al. 1993; Head et al. 2014) provide ample opportunity for gene flow, and the data we present here are consistent with panmixia. Interestingly, the levels of differentiation observed here are similar to those described for two species known to be panmictic, the Euro-

pean eel (*Anguilla anguilla*) ($F_{ST} \sim 0.0007$; Als et al. 2011; Pujolar et al. 2014) and American eel (*Anguilla rostrata*) ($F_{ST} \sim 0.00009$; Côté et al. 2013). However, it is important to keep in mind that while the life history characteristics of sablefish may provide one possible explanation for the apparent genetic homogeneity we observe, the study of differentiation among known spawning populations of sablefish is difficult because there is little data available on the specific location or time of spawning for this species. As a result, the sample collections made for this study were unable to target spawning groups of sablefish. This has the potential to influence the results of population genetic analyses in particular, as a priori hypotheses of population structure are restricted to where the fish were sampled and not potentially where they were spawned. Although it is important to recognize that while we cannot prove that panmixia exists by simply failing to reject our null hypothesis of genetic homogeneity (discussed in Gold and Richardson 1998; Heist and Gold 2000; Gold et al. 2001), the results we present here are suggestive of considerable mixing of fish between the areas sampled for this study outside of the reproductive season. While our study failed to uncover spatial genetic structure or signals of natural selection in the genome, further studies that investigate temporal changes in these population genetic parameters or studies that examine selection at early life stages may provide a more complete understanding of the genetic structure of this species.

Data accessibility

Sequence reads for this project have been deposited in the NCBI Sequence Read Archive (BioProject ID: PRJNA279314). The biological and environmental data for all samples submitted for sequencing, a fasta file containing sequences for the 2661 RAD tags generated by this study, and quality-filtered SNP genotypes for the 404 individuals included in this analysis as a vcf file are available in the online supplementary material¹.

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